



Department of Computer Science and Engineering

Academic Year 2023 – 2024 (odd Semester)

Degree, Semester & Branch: B.E., III & ECE-A

Course Code & Title: CS3353 & C Programming and Data Structures

Name of the Faculty member (s): Mrs. K.Jeyakarthika

Innovative Practice Description

Unit / Topic: Unit III / Conversion of infix to post-fix expression

Course Outcome: CO3

Topic Learning Outcome: TLO 9

Activity Chosen: Think Pair Share

Justification:

- Active learning strategy, in which students work on a problem posed by instructor, first individually (Think), then in pairs (Pair) or groups, and finally together with the entire class (Share).
- T (Think): Teacher asks a specific question about the topic. Students "think" about what they know or have learned, and come up with their own individual answer to the question. [Takes 5 Minutes].
- P (Pair): Teacher asks another question, related to the previous one that is suitable to deepen the students' understanding of the topic. Each student is paired with another student. They share their thinking with each other and proceed with the task. [Takes 5-10 Minutes].
- S (Share): Students share their thinking (or solution) with the entire class. Teacher moderates the discussion and highlights important points. [Takes 5-10 minutes].
- **Time Allotted for the Activity:** 20 minutes

Details of the Implementation:

- Think-Pair-Share innovative practice conducted for II year ECE A students, after explanation of the concept of infix, prefix and post fix and procedure to convert infix to postfix expression in classroom.
- First, I asked the students to think about the given expression for 5 minutes individually in the class (Think).
- Next, pair up or make a group with others in the class and discuss their points with them for 5 minutes. (Pair).
- Finally, I asked any two or three pair to explain the concept to the entire class. (Share).

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO9	PO10	PSO1
CO 2	3	2	2	2	2	1

(1 – Low 2 – Moderate 3 – High)**PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO9	PO10	PSO1
	3	2	1	2	2	1
Justification for correlation	Needs engineering and mathematical knowledge for Evaluating the expression concept	The given expression is analyzed and evaluated to find the result of the expression.	Students able to design solutions for expression using the evaluation concepts	Students individually think about the expression and also share their way to solve the expression with their team.	Able to explain the way of evaluation of expression with their team members.	Able to apply this application of stack concepts in emerging trends in Information Technology

- Images / Screenshot of the practice:**

***Figure:1 Glimpses of Think-Pair-Share Activity***

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- The students felt that this activity is very useful and got a clear idea about the concept of evaluation of expression.
- Working as a pair, they can share their knowledge with them and come up with better understanding about the topic.
- Also they can be able to apply it for other conversions like infix to prefix etc.

- ❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- From this activity, the students can get more clarity in the particular topic by discussing and sharing their views with the other students in the class.
- Coordination level among the students increased.
- Students enjoyed this learning activity and builds confidence with students.

- ❖ ***Challenges faced in implementation:***

Only a few student pairs were randomly selected to perform presentations in front of the class due to time constraints.

References:

- <https://www.theteachertoolkit.com/index.php/tool/think-pair-share>
- https://www.ritrjpm.ac.in/images/computer-science/2021-2022/KVS_CS8492_TPS.pdf
- https://www.ritrjpm.ac.in/images/computer-science/2022-2023/3_GM_CS3353_Evaluation_Expression_TPS.pdf
- <https://www.readingrockets.org/strategies/think-pair-share>